INCH-POUND

MIL-S-10057H(AR)
AMENDMENT 4
11 February 1993
SUPERSEDING
AMENDMENT 3
18 January 1991

MILITARY SPECIFICATION

SIMULATOR, HAND GRENADE, M116A1
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-S-10057H(AR), dated 6 January 1982, and is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2: Add the following under "DRAWINGS":

"INSPECTION EQUIPMENT DRAWINGS:

9200454 - Launcher".

3.4: Delete in in its entirety and substitute the following:

"3.4 Jumble. The simulator shall comply with the following requirements when tested as specified in 4.5.3:

This test is performed on first article samples only.

a. The simulator shall not function during test.
b. There shall be no loose pyrotechnic powder on the simulator or in the test box after jumbling.
c. The simulator components shall not separate during the test."

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* 3.5.2 Delay time. Delete "The delay time shall not be less than 6 seconds nor more than 11 seconds." and substitute "The delay time shall be not less than 6 seconds nor more than 12 seconds."

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Table I - First Article Inspection: Add after Disc and Fuse Assembly:

"Support, Priming Paste (Dwg. 12912921) Examination for defects	25	3.2	4.4.2.11"

PAGE 6

Table I - First article inspection Simulator Hand Grenade M116A1 Assembly (ALT). Delete in its entirety and substitute:

"Simulator Hand Grenade M116A1 Assembly (ALT) (Dwg. 9256467)			
Examination for Defects	50	3.2	4.4.2.7
Tests			
Jumble	10	3.4	4.5.3
Function @ Amb. Temp.	20 (a)	3.5	4.5.4.1
Function @ Cold Temp.	20 (b)	3.5	4.5.4.2

NOTES: Delete in their entirety and substitute:

"(a) Ten assemblies functioned in static and 10 in non-static phase.

(b) All assemblies functioned in the non-static phase."

PAGE 8

4.4.2.1 Disc., Delete major defect 202 in its entirety

PAGE 10

4.4.2.3:

Delete Major defect 102 and substitute:

"Major 102 Wall thickness (Note 1) 0.40% 3.2 Gage".

Add under "Notes":

"1. Wall thickness shall be gaged in non-overlap regions only".

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4.4.2.4 Disc and Fuse Assembly:

Major 101; Change to read: "Length from disc to end of fuse, prior to assembly to Priming Paste Support" (min.)

* Major 105; Change to read: "Priming paste or Ignition composition missing from Priming Paste Support"

Major 106; Change to read: "Hot Melt Glue fillet inadequate or less than 360°".

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* 4.4.2.5 Cover and Body Assembly: Major 104; Change to read: Priming paste or Ignition composition missing from, or damaged in Priming Paste Support".

PAGE 13

4.4.2.6: Delete Minor 201 in its entirety.

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4.4.2.9 Add: "Major 103, Instruction sheet missing, 0.40%, 3.2, Visual".

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4.4.2.10:

Major 101: Correct spelling of "contentsd" to "contents".

Delete Major 102 and Minor 201 in its entirety.

Add Major defect 103:

"Major 103 Strapping missing, broken or loose; 0.40%; 3.2; Visual/Manual

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Add Minor defect 206:

"Minor 206 Hardware improperly engaged; 0.65%; 3.2; Visual."

Add Minor defect 207: "Strapping improperly assembled; 0.65%; 3.2; Visual."

PAGE 17a

Add new paragraph 4.4.2.11 included.

PAGE 19

4.4.3.3 Delete in its entirety.

PAGE 20

4.4.3.4.1.1: *Delete paragraph in its entirety and substitute the following:*

"4.4.3.4.1.1 First three lots. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the requirements, 125 simulators shall be selected for this test. 75 simulator assemblies shall be functioned in the static phase and 50 simulators shall be functioned in the non-static phase. The lot shall be rejected if during the test, any of the following are exhibited in the combined 2 phases when tested as specified in 4.5.4"

a. One or more critical defects.

b. Six or more assemblies having major defects listed in Table V.

c. Average of all delay times for the combined phases of 4.4.3.4.1.1 and 4.4.3.4.2.1 is less than 8.5 seconds."

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Support, Priming Paste	1 OF 1	12912921
			NEXT HIGHER ASSEMBLY 9256463

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	I.D. at chamfered end	0.65%	3.2	Gage
202	Workmanship	1.00%	3.8	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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4.4.3.4.1.2: Delete in its entirety and substitute the following:

"4.4.3.4.1.2 Successive lot test. After three consecutive lots have complied with criteria of 4.4.3.4.1.1, 50 simulators shall be selected from each lot for this test. 25 simulators shall be functioned in the static phase and 25 simulators in the non-static phase (see 4.5.4). The lot shall be rejected if during the test, any of the following are exhibited when tested as specified in 4.5.4:

a. One or more critical defects.

b. Three or more assemblies having major defects listed in Table V.

c. Average of all delay times for the combined phases of 4.4.3.4.1.2 and 4.4.3.4.2.2 is less than 8.5 seconds".

4.4.3.4.2.1: Delete in its entirety and substitute the following:

"4.4.3.4.2.1 First three lots. Beginning with the first lot produced and continuing until 3 consecutive lots have complied with the acceptance criteria, 80 simulators shall be selected from each lot for test. All assemblies shall be functioned in the non-static phase. The lot shall be rejected if, during the test, any of the following is exhibited when tested as specified in 4.5.4:

a. One or more critical defects.

b. Four or more assemblies having major defects listed in Table V.

c. The average of all delay times for the combined phases of 4.4.3.4.1.1 and 4.4.3.4.2.1 is less than 8.5 seconds."

4.4.3.4.2.2: Delete in its entirety and substitute the following:

"4.4.3.4.2.2 Successive lot tests. After three consecutive lots have complied with the criteria of 4.4.3.4.2.1, 50 simulators shall be selected from each lot for test. All assemblies shall be functioned in the non-static phase. The lot shall be rejected if, during the test, any of the following is exhibited as specified in 4.5.4:

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a. One or more critical defects.

b. Three or more assemblies having major defects listed in Table V.

c. The average of all delay times for the combined phases of 4.4.3.4.1.2 and 4.4.3.4.2.2 is less than 8.5 seconds."

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4.5.3 Jumble.

Add: " This test performed on first article samples only."

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4.5.4: Delete in its entirety and substitute the following:

"4.5.4 Functioning.

4.5.4.1 Ambient temperatures (70 ± 5°F). The simulator assemblies shall be conditioned for 24 hours at the specified ambient temperatures. At the end of the conditioning period, the assemblies shall be tested in the following manner within 30 minutes after removal from the conditioning chamber.

4.5.4.1.1 Static. The assemblies, conditioned at ambient temperature, shall be mounted horizontally on a rigid horizontal support and fired by means of a lanyard attached to the fuse igniter cord. The assembly shall not be shaken or agitated in any manner other than normal handling within two (2) hours prior to test. Measuring of sound level intensity shall be accomplished with impact (sound) Analyzers, Microphone Systems and Sound Level Meters and the pick-up of the sound level shall be located 75 plus or minus 0.5 feet from the simulator test fixture.

4.5.4.1.2 Non-static. The assemblies, conditioned at ambient temperature, shall be fired by placing the assembly in an ejection machine (See Dwg. 9200454). The fuse igniter cord shall be held in a fixed position. There shall be no slack evident in the igniter cord prior to initiation. Upon ejection of the assembly, the simulator shall be initiated.

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The throwing range of the ejection machine shall not be less than 45 feet and the ground shall be free of large rocks. The assembly shall be observed for compliance with the requirement specified. Sound level intensity need not be measured during this test.

4.5.4.2 Cold temperature (minus 65 + 5°F). The simulator assemblies shall be conditioned for 16 hours at the required cold temperature and fired within 5 minutes after removal from the conditioning chamber by the same method specified in 4.5.4.1.2 (non-static). The functioning shall be observed for compliance with the requirements."

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6.5: Delete in its entirety.

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